Rethinking Long-Tail for Image Aesthetics Assessment

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Abstract

Real-world datasets often exhibit long-tail distributions, compromising the generalization and fairness of learning-based models. This issue is particularly pronounced in Image Aesthetics Assessment (IAA) tasks, where such imbalance is difficult to mitigate due to the severe long-tail distribution across both aesthetic labels and image scenes, as well as the ambiguity and uncontrollability of aesthetics. To address these issues, we introduce ELTA, an Enhancer Framework against Long-Tail for Aesthetics-oriented models, comprising ELTA 1.0 and its advanced extension ELTA 2.0. 1) ELTA 1.0 focuses on mitigating label-level imbalance. It enriches the representations of minority aesthetic labels via a dedicated mixup strategy, aligns features with their intrinsic aesthetic labels through a similarity consistency approach, and refines prediction distributions to improve pseudo-label quality. 2) ELTA 2.0 builds upon ELTA 1.0 and further tackles the more challenging scene-level long-tail imbalance. It automatically identifies long-tail scenes in the dataset, and then adopts an aesthetics-guided Text-to-Image (AG-T2I) model to generate new data with controlled scene content, image style and aesthetic quality, thus balancing scene distribution while preserving label distribution. Experiments on four representative datasets (AVA, AADB, TAD66K, and PARA) demonstrate that our proposed ELTA —particularly its full 2.0 version—achieves state-of-the-art performance by effectively mitigating the long-tail issue. To foster further research, we developed comprehensive benchmarks that evaluate 16 methods and released a supplementary dataset of over 50,000 images to alleviate imbalances in the existing dataset. To make these advancements fully accessible, we also provide a plug-and-play version of ELTA for easy integration with existing IAA methods, and a complete toolkit for long-tail analysis, automated data generation, and annotation. All resources will be available at here upon publication.

Keywords: Image Aesthetics Assessment, Long-Tail, Deep Learning

1 Introduction

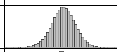
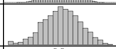
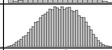
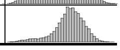
Benefiting from the growth and utilization of large-scale datasets, deep neural networks have achieved successes in computer vision tasks. However, these networks often face challenges in real-world applications due to data imbalance. It is commonly observed that models trained on long-tail data are biased towards majority groups with abundant samples, and away from minority groups with fewer samples. The bias compromises the model's generalizability and fairness, and raises concerns about ethical implications.

In some classical vision tasks, such as image classification and instance segmentation, various solutions [1–5] are proposed to mitigate the data imbalance. However, the progress of handling long-tail datasets in Image Aesthetics Assessment (IAA) lags behind classical vision tasks. As a recently emerging research, IAA is becoming increasingly vital in various domains, such as computational photography, art design, and recommender systems [6–11]. Unfortunately, as shown in Fig. 1, IAA encounters a particularly severe long-tail issue **on both label and scene distributions**, which is further complicated by its specificity.

This paper addresses the long-tail issue in IAA by focusing on two key questions and their specific challenges, which are respectively tackled by the ELTA 1.0 and ELTA 2.0 versions of our framework:

*ELTA 1.0: How to augment **existing data** while ensuring the accuracy of pseudo-labels?*

Challenge 1: Mismatch between Features and Labels. Image aesthetics tend to be sensitive to some augmentations, traditional data augmentation in the image dimension [12–15] may alter aesthetics ***without changing labels***, resulting in augmented images

Dataset	(a) Label Ground-truth Range								Label Distribution
	Minority		Majority				Minority		
	[0, 2)	[2, 3)	[3, 4)	[4, 5)	[5, 6)	[6, 7)	[7, 8)	[8, 10]	
AVA	6	491	7K	60K	116K	43K	3K	46	
AADB	506	826	1K	2K	880	2K	967	640	
TAD66K	180	2K	7K	11K	12K	11K	7K	1K	
PARA	0	81	1K	2K	8K	13K	3K	114	

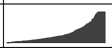
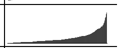
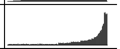
Dataset	(b) Scene Percentage Distribution								Scene Distribution
	Minority						Majority		
	[0, 3)	[3, 20)	[20, 35)	[35, 45)	[45, 55)	[55, 65)	[65, 75)	[75, 100]	
AVA	105	212	47	17	4	3	4	5	
AADB	88	252	37	5	3	2	0	1	
TAD66K	104	264	14	2	2	0	0	2	
PARA	109	249	17	7	5	5	1	3	

Fig. 1: The long-tail issue affects both label and scene distributions in IAA datasets. (a) The scores of all dataset labels are normalized to the range [0, 10]. (b) The number of samples in each scene is calculated as a percentage relative to the scene with the most samples, and then the scene distribution is analyzed based on these percentages.

with incorrect labels. Therefore, augmentation in the feature dimension, using existing labels to compute pseudo-labels, offers a potential solution. On the other hand, the inherent ambiguity of IAA datasets may lead to "noise". For example, the same image may contain different opinions, all of which are treated as ground truth. This ambiguity significantly increases the learning difficulty of models and leads to severe mismatches between labels and learning features (Fig. 2).

Challenge 2: Lack of Well-defined Categories. Compared to the long-tail issue encountered in classification tasks, the IAA tasks present unique challenges. The subjectivity in the labeling process and the ambiguity between different aesthetic criteria levels make it difficult to define clear categories. Creating artificial boundaries based on scores in IAA is both ambiguous and not universally applicable, and this absence of well-defined categories hinders the use of many class-based methods to balance label distribution, such as re-sampling [16–19], re-weighting [20–24].

Challenge 3: Smoothness of Predicted Distributions. Recent semi-supervised learning methods [25–27] rely heavily on model confidence and specific threshold to identify reliable pseudo-labels. Unlike typical models, the output logits of IAA models frequently exhibit smoother output distributions and ambiguity, where confident and uncertain predictions blend seamlessly. Consequently, simply filtering out uncertain predictions by a high threshold becomes ineffective for generating accurate pseudo-labels in IAA tasks.

*ELTA 2.0: How to generate **new data** while balance the distribution of pseudo-labels?*

Challenge 4: Style Consistency in Generated Data. Augmenting existing data, while beneficial, is inherently limited by the richness and diversity of the original dataset, which poses a significant bottleneck in directly addressing the long-tail scene distribution. Consequently, generating new data becomes a more direct approach to tackle the scene imbalance. IAA datasets are often sourced from photography or social media platforms, where images reflect specific temporal, cultural, or platform-specific trends. Consequently, images from different datasets may exhibit significant stylistic disparities, even within similar scenes (Fig. 3). Introducing samples with mismatched styles risks creating a distributional shift, where the generated data fails to represent the minority aesthetics of the target dataset. This misalignment can confuse the model, leading to degraded generalization and inaccurate pseudo-labels. Therefore, the challenge lies in generating new data that not only augments the minority samples to balance the long-tail distribution but also adheres to the specific stylistic and contextual demands of the target IAA dataset.

Challenge 5: Preserving Label Distribution Integrity. Introducing new data to mitigate the long-tail effect in scene distributions can inadvertently exacerbate imbalances in the label distribution. Newly generated samples, especially those targeting underrepresented scenes or aesthetic qualities, may skew the overall label distribution if their aesthetic scores are not carefully controlled. Manual curation of new data to ensure both scene diversity and balanced aesthetic quality is labor-intensive and impractical at scale, particularly given the subjective nature of aesthetics. Thus, the challenge is to develop a data generation strategy that simultaneously ensures controllable aesthetic quality and scene diversity, and preserves or enhances the balance of the label distribution to avoid amplifying existing long-tail issue.

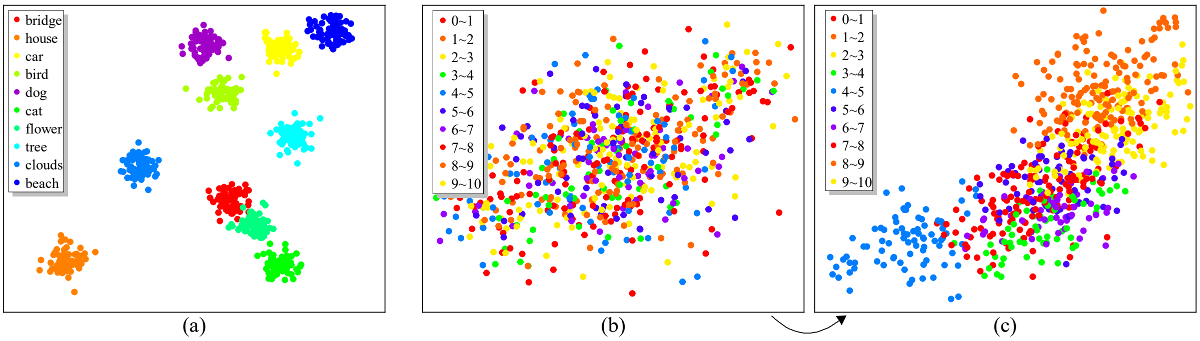


Fig. 2: T-SNE visualization of feature distributions extracted by (a) a classification network, (b) IAA network, and (c) our ELTA-aligned features, respectively. The entire IAA score range is divided into 10 segments, with feature points in each segment marked in different colors. The classification features (a) clearly distinguish between different categories, while the IAA features (b) exhibit noticeable confusion. (c) After alignment with our ELTA framework, this confusion is effectively mitigated.

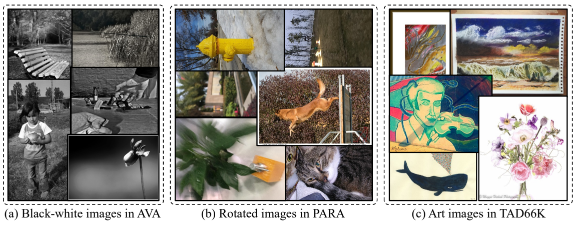


Fig. 3: The IAA dataset images exhibit some stylistic differences. For instance, (a) the AVA dataset [28], comprising photos taken before 2012, include many black-and-white images. (b) The PARA dataset [29], designed for personalized IAA, contains numerous casually captured and rotated images. (c) The TAD66K dataset [30] features images depicting various scenes rendered in an artistic painting style.

In addition to the challenges outlined above, research on long-tail issue in IAA faces the following limitations:

Limitation 1: Completeness and Fairness. Commonly used metrics in IAA model evaluation, such as Pearson Linear Correlation Coefficient (PLCC) and Spearman Rank Correlation Coefficient (SRCC), are designed to assess the holistic performance of models. However, these metrics are insufficient for evaluating a model's robustness to long-tail data distributions. Furthermore, there is a lack of standardized, fair benchmarks to systematically compare the effectiveness of different approaches in addressing the long-tail issue. This gap hinders the ability to objectively evaluate and improve models for long-tail IAA tasks.

Limitation 2: Applicability. Due to the pervasive influence of long-tail distributions in all IAA datasets, the long-tail issue is a fundamental challenge in IAA research. However, the research community lacks practical tools directly applicable to IAA. Additionally, there is no comprehensive framework to systematically mitigate the long-tail issue across the entire pipeline, from data collection and augmentation to model training and evaluation. This absence of robust tools and frameworks limits progress in effectively addressing the long-tail issue in IAA.

To address the aforementioned challenges and limitations, we argue that developing effective methods to mitigate the long-tail issue, along with defining suitable evaluation metrics, and then providing a complete, unified benchmark and tool, is highly desired. As such, we make four distinct contributions in this work:

- **Insight.** We are the first to systematically identify and analyze the dual long-tail issue in IAA, which manifests in both label and scene distributions. We reveal its severe negative effects on model performance and its adverse implications for fairness in real-world applications. Furthermore, we articulate the unique set of challenges that arise when addressing long-tail issue from an IAA-specific perspective.
- **Methodology.** We propose ELTA, a novel method that tackles data imbalance at two levels. At the label level, ELTA augments minority features, aligns features more closely with their corresponding labels, and improves the accuracy of pseudo-labeling. At the scene level, ELTA employs an aesthetics-guided Text-to-Image (AG-T2I) model to generate new data. This process ensures control over scene content, image style, and aesthetic quality, effectively balancing the scene distribution while preserving label distribution. To our knowledge, ELTA is the first solution developed specifically for the long-tail IAA problem and achieves state-of-the-art performance on four representative IAA datasets.
- **Benchmark.** We establish the most extensive benchmark for long-tail IAA to date, featuring a thorough evaluation of 16 baseline methods (Table 2). Besides, we will also maintain an online benchmark to dynamically trace the development of this field. Our study serves as a potential catalyst for promoting

Table 1: Comparison of ELTA 1.0 and ELTA 2.0.

#	Level	Motivation	Component	Benchmark	Resource	Pub. Year
ELTA 1.0	Label	Augment Existing Data via Features	Mixup, Feature Align, Distribution Shaping	9 methods on 1 Level	3 Plug-and-play Modules	ICML 2024
ELTA 2.0	Scene	Generate New Data by T2I Model	Aesthetic-Guided T2I, Stylistic Align	13 methods on 2 Levels	Analysis and Dataset Tools, 50,000 Supplemental Images	IJCV 2025

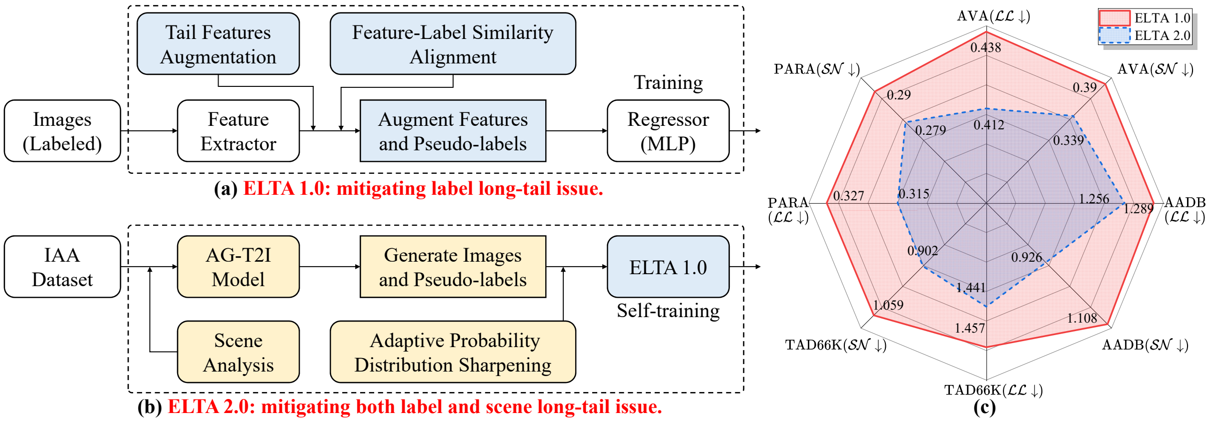


Fig. 4: Architecture of the proposed ELTA framework and comparative performance. (a) ELTA 1.0 is trained to balance the label distribution by augmenting features (Sec. 4). (b) ELTA 2.0 integrates AG-T2I with ELTA 1.0 to synthesize minority images, thereby addressing long-tail issue at both the label and the scene level (Sec. 5). (c) On four representative IAA datasets, ELTA 2.0 achieves substantially better performance than ELTA 1.0 on minority samples (lower is better; see Sec. 6.1 for metrics).

large-scale model comparisons and advancing future long-tail IAA research.

- **Toolkit and Resources.** We provide the community with a suite of practical tools. ELTA not only provides plug-and-play modules that can be seamlessly integrated with existing approaches, but also includes a complete toolkit for automated long-tail analysis, data generation, and annotation for IAA datasets. Moreover, we contribute a curated collection of over 50,000 images designed to alleviate the long-tail problems present in existing datasets.

This work extends our previous conference version [31] (called ELTA 1.0 in this paper). The extension is as follows (refer to Table 1 and Fig. 4). **1) Enhanced Analysis and Methods:** We provide a more comprehensive analysis of the long-tail issue in IAA, with a particular focus on the scene-level long-tail distribution, which was not fully addressed in our earlier work. We identify the unique challenges inherent to IAA tasks and propose a novel perspective utilizing an aesthetics-guided T2I model to effectively mitigate scene-level imbalances. **2) Expanded Benchmark:** We establish a more extensive benchmark, incorporating a broader range of methods, additional evaluation levels, and nearly double the number of experiments compared to our previous work. Additionally, we contribute a curated collection of over 50,000 supplemental images, analysis and dataset tools to further alleviate the long-tail issue in existing IAA datasets. **3) Improved Presentation and Organization:** We have made many efforts to improve the presentation (e.g., dataset, framework, key results) and organization of our paper. Besides, several sections have been rewritten to improve the readability and provide more detailed explanations about the introduction. **4) Future Research Directions:** Drawing on our benchmarking results, we highlight several fundamental research directions and challenges, providing a roadmap for future advancements in this field.

2 Related Works

2.1 Image Aesthetics Assessment

General IAA encompasses three types of tasks: aesthetic binary classification (aesthetically positive or negative) [32, 33], aesthetic score regression [30, 34–36] and score distribution prediction [37–42], while personalized IAA adopts an aesthetic model for an individual user's preferences [29, 43, 44]. Traditional methods rely on manually designed and extracted features from images. These visual features are then mapped to annotated labels via classifiers or regressors. In contrast, learning-based methods use Convolutional Neural Networks (CNNs) or Vision Transformers (ViTs) with robust feature extraction capabilities to replace the manual process.

Multimodal Large Language Models (MLLMs), such as LLaVA [45, 46], mPLUG-Owl [47, 48], and the QwenVL [49, 50] series have shown remarkable performance in image captioning and demonstrated potential in understanding high-level visual content [51]. Recent studies also show their ability to perceive low-level visual attributes [52, 53]. Building on these strengths, recent efforts [53–57] have applied these models to IAA, adapting them for aesthetic evaluation tasks. However, most MLLMs [54, 57] treat scores as plain text and tokenize each complete score into multiple discrete tokens, making them insensitive to the numerical values. As a result, they struggle to perceive the long-tailed distribution of scores.

While existing learning-based IAA methods have achieved impressive results, they confront a critical issue: model bias caused by the long-tail distribution of the datasets [28–30, 58–60]. Our analysis reveals that this issue is particularly severe in IAA, manifesting as a dual long-tail problem across both label and scene distributions (Fig. 1). This dual imbalance leads to models with poor generalization, questionable fairness, and a bias towards majority classes (i.e., mid-range scores and common scenes).

This long-tail label problem was first highlighted by He *et al.* [30] in their construction of the TAD66K dataset, where they employed a dedicated balancing strategy to mitigate it. However, a fundamental hurdle lies in the inherent nature of human annotation itself. The inclination of annotators to shy away from extreme ratings inevitably induces a long-tail distribution, regardless of the dataset construction. The scene-level imbalance presents an even greater challenge, as rectifying it through data generation risks introducing stylistic inconsistencies or inadvertently exacerbating the label imbalance. To systematically address this dual challenge, our work pioneers a two-level framework named ELTA. At the label level, ELTA enhances minority representation by augmenting features and enforcing feature-label similarity alignment. At the scene level, it employs a novel aesthetics-guided Text-to-Image model to generate new, stylistically consistent data that simultaneously balances the scene distribution while preserving label integrity.

2.2 Long-tail Learning

Previous research on long-tail learning has predominantly focused on classification tasks. ***Typical methods*** include re-sampling [16–19, 65] and re-weighting [20–24], which rebalance the contribution of each class. However, these methods are less applicable to aesthetics-oriented scoring tasks, which have no explicit class boundaries. ***Transfer learning methods*** [5, 66–69] aim to transfer knowledge from majority classes to enhance the learning of the minority. However, the highly abstract nature of aesthetics often hinders effective knowledge extraction and transfer. Some other methods use data augmentation techniques to expand datasets [12–15], but they often risk unintentionally degrading aesthetic quality and introducing inconsistencies between the original labels and the augmented images.

Although a few recent studies have focused on ***long-tail regression tasks*** [63, 64], we find that these methods have not fully overcome the aforementioned challenges and their performance gains on IAA tasks are relatively limited. This limitation may stem from the methods' inadequate consideration of the IAA task's inherent complexities, which include subjectivity of aesthetic perception and the uncertainty of aesthetic labels. Instead, we fully consider these aesthetic properties and design a dedicated solution. To further advance progress in this domain, we also contribute a comprehensive toolkit and the largest benchmark, providing the IAA community with the necessary resources to systematically evaluate and address the long-tail issue.

3 Preliminaries

3.1 Problem Setup

We consider two primary tasks, distribution prediction and score regression, based on whether the labels are in distribution or single-valued form. **1)** In the distribution prediction task, the training set with N samples is denoted as $\boldsymbol{D} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}_{i=1}^{N}$. Here, $\boldsymbol{x}_i$ represents an input image, and the corresponding ground truth distribution of human ratings is indicated by the empirical probability mass function (PMF) $\boldsymbol{y}_i = [y_i^1, y_i^2, ..., y_i^C]$. Each element in $\boldsymbol{y}_i$ represents an aesthetic quality level, with C denoting the total number of these levels. The goal of model training is to predict the PMF $\hat{\boldsymbol{y}}_i = [\hat{y}_i^1, \hat{y}_i^2, ..., \hat{y}_i^C]$, which should be an accurate estimate of $\boldsymbol{y}_i$. For each image $\boldsymbol{x}_i$, the ground truth Mean Opinion Score (MOS) and the predicted score can be calculated as $s_i = \sum_{j=1}^{C} j \times y_i^j$ and $\hat{s}_i = \sum_{j=1}^{C} j \times \hat{y}_i^j$, respectively. **2)** In the score regression task, each image $\boldsymbol{x}_i$ is assigned a scalar score y_i. Despite this, the model can still output a distribution and then minimize the gap between the score $\hat{s}_i$ derived from the distribution and the ground truth label s_i.

3.2 Boundary Definitions

Our primary focus will be on addressing these amplified errors observed in minority label and scene samples. Thus, before delving into the solution, it's essential to establish precise definitions and boundaries for the majority and minority.

Label Boundaries. IAA datasets inevitably exhibit long-tail distributions, where a large number of images' aesthetic qualities cluster around the mean $\bar{s} = \frac{1}{N}\sum_{i=1}^{N} s_i$. This leads to a tendency for the models to output medium scores, which means the minority

Table 2: Summary of 16 models on our benchmarks. We count 11 SOTA IAA models with publicly available official code. Additionally, we have retrained 5 models of other vision tasks (indicated by '*') that demonstrated outstanding performance on IAA tasks or long-tail tasks.

	No.	Model	Years	Pub.	Method	Code
CNN	1	NIMA [37]	2018	TIP	score distribution predicting CNN	Tensorflow
	2	BIAA [61]	2019	TCYB	meta-learning, bilevel optimization	PyTorch
	3	HGCN [39]	2021	CVPR	graph convolution networks	Jittor
	4	TANet [30]	2022	IJCAI	attention, adaptive features	PyTorch
Transformer	5*	Swin [62]	2021	CVPR	shifted windows transformer	PyTorch
	6	MUSIQ [40]	2021	ICCV	multi-scale image quality transformer	Tensorflow
	7	MaxViT [41]	2022	ECCV	multi-axis vision transformer	Tensorflow
	8	EAT [42]	2023	ACMMM	deformable transformer	PyTorch
DIR	9*	L&FDS [63]	2021	ICML	intensity-guided consistent feature	PyTorch
	10*	RankSim [64]	2022	ICML	Ranking similarity regularization	PyTorch
MLLMs	11*	LLaVA [46]	2024	CVPR	visual instruction tuning	PyTorch
	12*	Q-Instruct [53]	2024	CVPR	low-level visual perception	PyTorch
	13	AesExpert [55]	2024	ACMMM	aesthetic multi-modality instruction tuning	PyTorch
	14	Q-Align [57]	2024	ICML	instructing LMMs with text-defined levels	PyTorch
Ours	15	ELTA 1.0 [31]	2024	ICML	against long-Tail by feature augment	PyTorch
	16	ELTA 2.0	2025	IJCV	against long-Tail by feature augment and T2I model	PyTorch

samples with labels that deviate substantially from the mean are prone to larger prediction errors, as indicated by $|\hat{s}_i - s_i| \propto |s_i - \bar{s}|$. To define the minority label, we sort the dataset by aesthetic score labels and designate the highest 20% and the lowest 20% of scores as the minority label, which will be the primary focus of our analysis and improvement efforts. Statistically, this threshold aligns with common practices in long-tail learning, where the top and bottom quintiles are frequently used to define minority classes, as they balance the trade-off between capturing the tail's diversity and maintaining adequate sample sizes.

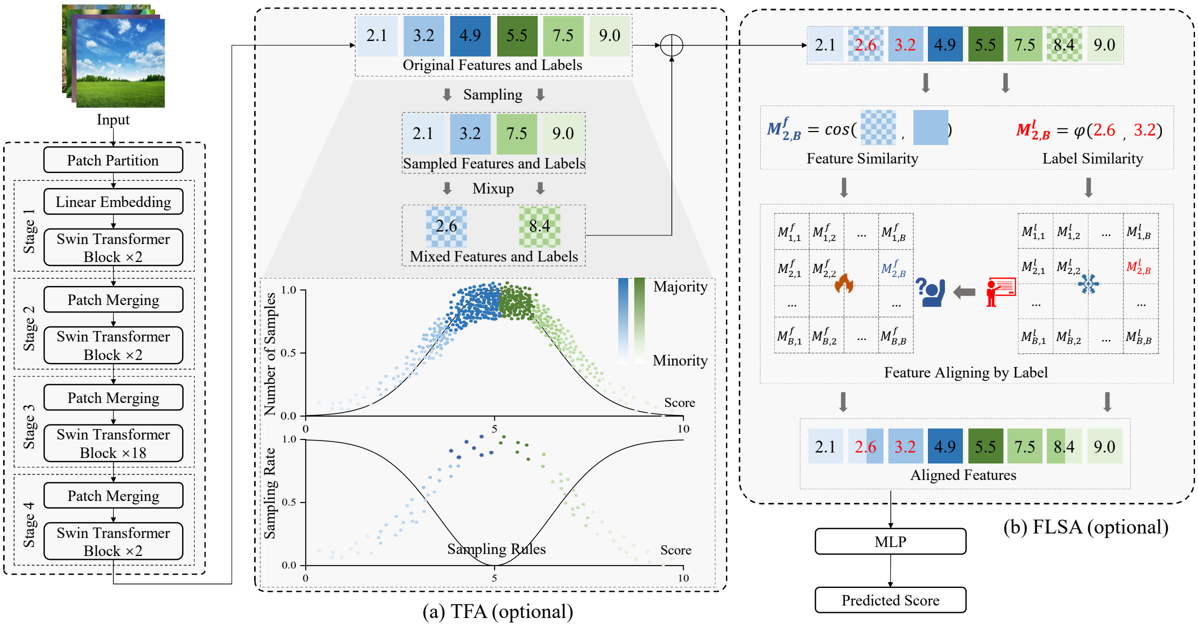
Scene Boundaries. To define the minority scenes, we first compute the frequency of each scene category based on the inferred scene labels derived from pre-trained scene classification models [70]. To analyze the distribution of scenes, we first calculate the percentage of samples in each scene relative to the most sampled scene. After sorting these percentages, we initially designate the 20% of scenes with the lowest sample counts as minority scenes. Therefore, as illustrated in Fig. 1(b), scenes with a percentage between [0,20) are considered minority scenes in this study.

4 ELTA 1.0: Augment Existing Data

The overall structure of ELTA is shown in Fig. 4. In this section, we discuss how to augment existing data in feature dimensions to solve the long-tail issue of labels. ***First***, unlike traditional pixel-dimension augmentation methods, our ELTA implements feature-dimension augmentation based on the mixup [71] to augment the minority. We propose an improved sampling strategy to assign different sampling weights to the minority and majority, enabling efficient generation of minority features even in the absence of well-defined categories. ***Second***, Gong et al. [64] state that a well-trained model should exhibit a key characteristic: items closer in label space should also be nearer in feature space. Inspired by this idea, a Feature-Label Similarity Alignment module is proposed to maximize the consistency between feature similarity and label similarity. Specifically, we take similarity as a metric for measuring distance, leverage label similarity to refine feature similarity, and optimize the model's representation learning. The pipeline is shown in Fig. 5.

4.1 Tail Features Augmentation

Image augmentation approaches such as cropping, flipping, and rotation, which directly manipulate the original image, can potentially destroy the intricate aesthetic information, creating an irreversible impact independent of the network's learning process [38, 72]. Therefore, feature-dimension enhancement is a more suitable augmentation approach for IAA ***without altering the original label***. By operating on higher-dimensional features extracted from images, rather than modifying raw pixels, it preserves essential aesthetic qualities without changing the label.



(a) TFA (optional)

(b) FLSA (optional)

Fig. 5: To augment existing data in feature dimension, ELTA consists of two **optional** components: (a) TFA performs mixup between instances sampled by our dedicated strategy to enhance minority features; (b) FLSA aligns features and labels based on their similarity consistency. The Swin Transformer V2 [62] is selected as the default network backbone.

As a classic technique, mixup [71] can be applied to generate new samples and their associated labels through linear interpolation. Specifically, considering an image-label pair $(\boldsymbol{x}_i, \boldsymbol{y}_i)$ and $(\boldsymbol{x}_j, \boldsymbol{y}_j)$, let $\boldsymbol{z}_i^k, \boldsymbol{z}_j^k = f^k(\boldsymbol{x}_i), f^k(\boldsymbol{x}_j)$ be the features extracted at the k-th layer of a deep learning model from $\boldsymbol{x}_i, \boldsymbol{x}_j$. Then the mixed feature $\tilde{\boldsymbol{z}}^k$ and label $\tilde{\boldsymbol{y}}$ are generated as follows:

$$\begin{aligned} \tilde{\boldsymbol{z}}^k &= \lambda \boldsymbol{z}_i^k + (1-\lambda)\boldsymbol{z}_j^k, \\ \tilde{\boldsymbol{y}} &= \lambda \boldsymbol{y}_i + (1-\lambda)\boldsymbol{y}_j. \end{aligned} \tag{1}$$

It should be noted that when $k = 0$, the input feature $\boldsymbol{z}_i^0$ is identical to the original image $\boldsymbol{x}_i$, reverting to the initial version of mixup. In the following text, the superscript k will be omitted for brevity.

A mixup operation involves choosing two samples, and most of the previous work used random sampling. However, while random sampling may be effective in class-balanced datasets, it proves inadequate for long-tail IAA datasets. This is due to the inherent dominance of majority instances within such datasets, resulting in an unintended abundance of majority features when employing random sampling and deviating from our original objective of enhancing minority representation.

To solve the above problem, we use the label information to guide more sampling of the minority and design a two-stage strategy. Initially, the sampling probability $P(i)$ for $\boldsymbol{z}_i$ is calculated based on its score deviation from the mean score of all samples. Samples far from the mean have high sampling probabilities, thus ensuring a bias towards the minority. Once the first sample feature $\boldsymbol{z}_i$ is chosen, the probability $P(j|i)$ that each of the remaining features is paired with $\boldsymbol{z}_i$ can be calculated. This probability hinges on the score difference between the two samples. A smaller difference means a higher chance of pairing, thus ensuring the aesthetic similarity within pairs. The equations are detailed below:

$$P(i) = \frac{\exp\left(|\bar{s} - s_i|/\tau_1\right)}{\sum_{k=1}^{B} \exp\left(|\bar{s} - s_k|/\tau_1\right)}, \tag{2}$$

$$P(j|i) = \frac{\exp\left(\tau_2/|s_i - s_j|\right)}{\sum_{k=1,2,\cdots,i-1,i+1,\cdots,B} \exp\left(\tau_2/|s_i - s_k|\right)}, \tag{3}$$

where $\bar{s}$ represents the mean score of all the samples within that batch, and B is the batch size. τ_1 and τ_2 are temperature hyper-parameters that can be used to regulate the level of preference for the minority samples.

In traditional methods, the mixing factor λ in Eq. (1) is randomly selected from a beta distribution. To enhance the representation of minority data, we change λ to be obtained from the following equation:

$$\begin{aligned} \lambda &= \frac{P(i)}{P(i) + P(j)} \\ &= \frac{\exp(|\bar{s} - s_i|/\tau_1)}{\exp(|\bar{s} - s_i|/\tau_1) + \exp(|\bar{s} - s_j|/\tau_1)}. \end{aligned} \tag{4}$$

where $P(j)$ is computed by Eq. (2). This allocation of weights allows the sample which is more towards the minority in the pair to have a dominant influence in the mixup process, reducing the impact of accidentally sampling a majority sample.

4.2 Feature-Label Similarity Alignment

In IAA tasks, features often exhibit significant confusion, as illustrated in Fig. 2. This presents a challenge when addressing data imbalance through feature augmentation. Previous work [64] has shown items closer in label space should also be nearer in feature space. By leveraging this insight, label similarity can guide the refinement of features. This alignment process fosters a harmonious relationship between features and labels, promoting consistency and rectifying potential mismatches.

Specifically, the images $\boldsymbol{x}$ are first fed into the encoder, which extracts feature vectors $\boldsymbol{z}$ that capture their essential characteristics. These feature vectors are then compared pairwise using cosine similarity and yield a feature similarity matrix, denoted as $M^f \in \mathbb{R}^{B\times B}$, where the element of M^f can be formulated as:

$$M_{i,j}^f = \cos(\boldsymbol{z}_i, \boldsymbol{z}_j) = \frac{\boldsymbol{z}_i \cdot \boldsymbol{z}_j}{|\boldsymbol{z}_i| \cdot |\boldsymbol{z}_j|}. \tag{5}$$

Each element within this matrix holds a value that quantifies the similarity between a specific pair of feature vectors. Besides, the similarity matrix of labels $M^l \in \mathbb{R}^{B\times B}$ can be expressed as a function φ derived from their absolute differences:

$$M_{i,j}^l = \varphi(s_i, s_j) = 1 - \frac{|s_i - s_j|}{\max\limits_k s_k - \min\limits_k s_k}. \tag{6}$$

In particular, within the label similarity matrix, $M_{i,j}^l = 1$ signifies corresponding sample pairs that share identical labels, representing the highest level of similarity. Conversely, $M_{i,j}^l = 0$ indicates a complete absence of relevance.

As features are adaptable through learning, the backpropagation process utilizes the label similarity matrix to guide the feature similarity matrix toward closer alignment, effectively minimizing the discrepancy between them. The alignment loss can be expressed as $\mathrm{MSE}(M^f, M^l)$. A tunable parameter, which controls the weight of this loss, is then multiplied with it. Subsequently, this product is added to the original supervised loss (e.g., EMD), acting as a regularization term.

5 ELTA 2.0: Generate New Data

Training robust IAA models is challenging due to the long-tail label distribution and imbalanced scene distribution in existing datasets. This scene imbalance leads to unreliable model prediction on less common scene types. To address this issue, for model self-training, ELTA 2.0 employs AG-T2I model to generate new data with controlled scene content and aesthetic quality, balancing scene distribution without affecting label distribution. Moreover, we introduce an Adaptive Probability Distribution Sharpening module to alleviate the smoothness of predicted distributions and obtain relatively accurate pseudo-labels.

Stage I: Baseline T2I Model Pre-training

To establish a robust T2I baseline with general image generation capabilities, we first pre-train a diffusion-based T2I model [73] (named UltraEdit) on the large-scale ULTRAEDIT dataset [73]. We adopt the editing diffusion model introduced in InstructPix2Pix [74], which uses Stable Diffusion v1.5 [75] as the backbone diffusion model. After pre-training, the baseline model supports instruction- and region-based editing.

Stage II: Stylistic Alignment and Aesthetics-Guided Fine-tuning

A critical challenge in generating data for model self-training is ensuring that the newly introduced data does not introduce stylistic inconsistencies relative to the training IAA datasets. Moreover, it is important that the new data does not exacerbate existing long-tail issue in the label distribution. To address these concerns, the second stage aims to enable the model to generate samples that are both stylistically coherent with the target IAA dataset and achieve controlled aesthetic scores. For each target IAA dataset, the data preparation and fine-tuning process of the baseline T2I model is as follows (Fig. 6).

Target and Source Data Preparation. For the target IAA dataset, we extract images denoted as I_{target}, with the corresponding ground-truth aesthetic score S_{target}. Then, a scene classification model [70] obtains a corresponding scene caption T_{target} for each I_{target}. This process produces a collection of target tuples $\mathcal{D}_{\text{target}} = \{(I_{\text{target},j}, T_{\text{target},j}, S_{\text{target},j})\}_{j=1}^{N_{\text{target}}}$. For each target tuple $(I_{\text{target},j}, T_{\text{target},j}, S_{\text{target},j})$ in $\mathcal{D}_{\text{target}}$, we randomly sample an editing instruction T_e from the ULTRAEDIT dataset's instruction pool. Using T_e, we apply an Image-to-Image (I2I) model [76] to edit $I_{\text{target},j}$, generating a corresponding source image $I_{\text{source},j}$.

The fine-tuning process of AG-T2I model aims to revert $I_{\text{source},j}$ to $I_{\text{target},j}$ by the conditioning information. This process, guided by target data, ensures generated images align closely with the target IAA dataset's style.

Aesthetic-Guided Fine-tuning. The core of the fine-tuning process involves iterative denoising steps. Let z_k be the noisy latent representation of $I_{\text{source},j}$ at diffusion timestep k. The aesthetic-guided denoising step aims to predict the noise to remove from z_k to approach the latent representation of $I_{\text{target},j}$, while ensuring the generation aligns with $S_{\text{target},j}$. This step operates as follows:

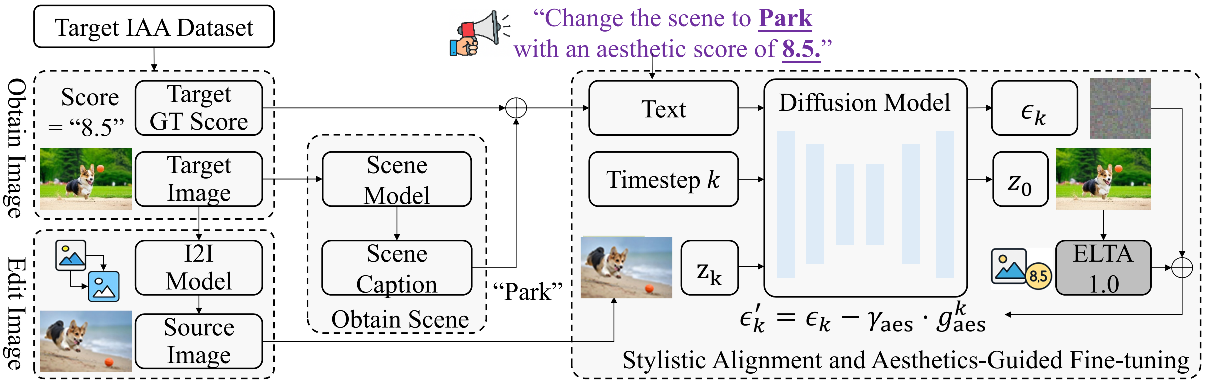


Fig. 6: Data preparation and aesthetic-guided fine-tuning process of AG-T2I model.

i. Standard Noise Prediction: The diffusion model ε_θ predicts the noise based on the current noisy latent z_k, and the conditioning information $(k, T_{\text{target},j}, S_{\text{target},j})$:

$$\epsilon_k = \varepsilon_\theta(z_k, k, T_{\text{target},j}, S_{\text{target},j}). \tag{7}$$

ii. Predict Target Latent: An estimate of the target latent, z_0, is obtained using the predicted noise ϵ_k:

$$z_0 = \frac{1}{\sqrt{\alpha}_k}(z_k - \frac{1-\alpha_k}{\sqrt{1-\bar{\alpha}_k}}\epsilon_k) \tag{8}$$

where $\bar{\alpha}_k$ is a noise schedule parameter at timestep k.

iii. Decode and Evaluate Aesthetics: The predicted target latent is decoded to an image $I_0 = \mathcal{G}(z_0)$, where $\mathcal{G}$ is the decoder. Its aesthetic score S_0 of I_0 is then computed by our ELTA 1.0 model.

iv. **Aesthetic Loss and Gradient**: An aesthetic loss is computed based on the difference between the predicted score and $S_{\text{target},j}$:

$$L^k_{\text{aes}} = (S_0 - S_{\text{target},j})^2. \tag{9}$$

The gradient of this loss with respect to the current noisy latent z_k provides the guidance signal:

$$g^k_{\text{aes}} = \nabla_{z_k} L^k_{\text{aes}}. \tag{10}$$

v. **Corrected Noise Prediction**: The initial noise prediction is adjusted using the aesthetic gradient:

$$\epsilon'_k = \epsilon_k - \gamma_{\text{aes}} \cdot g^k_{\text{aes}}, \tag{11}$$

where γ_{aes} is a scalar controlling the strength of the aesthetic guidance.

vi. **Denoising Step**: The latent z_{k-1} for the next diffusion step is derived using the corrected noise prediction ϵ'_k.

This fine-tuning process, incorporating aesthetic guidance, enables the AG-T2I model model to generate images that not only stylistically align with $T_{\text{target},j}$ but also meet the specific aesthetic quality defined by $S_{\text{target},j}$.

Stage III: Targeted Data Generation for Self-Training

Stage III uses the fine-tuned AG-T2I model model from Stage II to create synthetic samples with controlled aesthetic scores. The process is outlined below.

Identifying Minority Scenes and Anchor Aesthetic Scores. We first identify minority scene categories (Sec. 3.2), denoted as $SE = se_1, se_2, \ldots, se_N$, by analyzing the scene distribution in the target IAA dataset, where N represents the total number of these categories. Next, we define anchor aesthetic scores, $SC = sc_1, sc_2, \ldots, sc_M$, where M is the number of discrete scores chosen. Since aesthetic scores are continuous (e.g., [0, 10]), traversing all values is impractical. Instead, we select integer values, such as $SC = 0, 1, 2, \ldots, 10$, as anchor points for targeted generation.

Instruction Formulation and Image Generation. For each minority scene category $se_k \in SE$ and anchor aesthetic score $sc_m \in SC$, we follow these steps.

i. Randomly select an input image from the target IAA dataset.

ii. Create an editing instruction, T_{instr}, to guide the AG-T2I model. The instruction is formatted as: "Change the scene to se_k with an aesthetic score of sc_m."

iii. Feed the input image and instruction into the fine-tuned AG-T2I model from Stage II to generate a synthetic image depicting scene se_k with aesthetic quality matching score sc_m.

This process is repeated for R rounds to ensure that samples for each anchor aesthetic score are generated uniformly, to avoid amplifying the existing long-tail distribution. Since minority samples have fewer samples than majority classes, uniform sampling increases the overall proportion of minority samples. The T2I model's randomness ensures diverse images

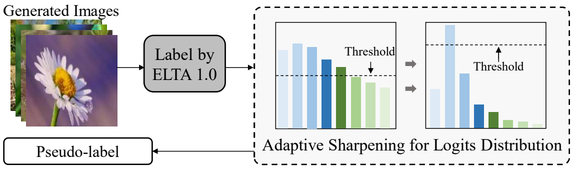


Fig. 7: APDS reshapes the predicted distributions to work together with high thresholding for high-quality pseudo-label selection.

are generated, even with identical instructions. These synthetic images are added to the IAA model's self-training phases, addressing the imbalance in minority scenes and enhancing the model's robustness and generalization.

Stage IV: High-quality Minority Samples Selecting

While expanding the minority from unlabeled samples offers a promising approach, how to properly utilize these samples requires careful consideration. We employ an Adaptive Probability Distribution Sharpening strategy to pseudo-label unlabeled data, specifically seeking minority samples to enrich diversity (Fig. 7).

Challenges in Selecting High-quality Minority Samples. Existing research [25–27] demonstrates that setting a high threshold and retaining only predictions with confidence exceeding this threshold can effectively filter out samples with large prediction errors. However, this approach confronts challenges due to the subjective and continuous nature of IAA labels, which leads to a softer, smoother, and less distinct predicted distribution. Consequently, identifying an optimal threshold for selecting high-quality minority samples is difficult. Especially, setting the threshold too high may exclude almost all potential samples, whereas a too-low threshold could include numerous samples with significant errors.

Reshaping the Distribution Adaptively. To overcome the challenges posed by IAA's smooth predicted distribution, we propose a temperature scaling technique inspired by [77]. By adjusting the original distribution, we aim to facilitate effective threshold selection for identifying reliable pseudo-labels. Specifically, given the logit vector $\boldsymbol{z}_i$, the new prediction confidence distribution $\hat{\boldsymbol{y}}_i$ is obtained as follows:

$$\hat{\boldsymbol{y}}_i = \mathrm{Softmax}(\boldsymbol{z}_i/\tau) = \frac{e^{\boldsymbol{z}_i/\tau}}{\sum_{j=1}^{B} e^{\boldsymbol{z}_j/\tau}}, \tag{12}$$

where τ is a temperature parameter that scales logit vectors. Increasing τ above 1 smooths the probability distribution, while below 1 sharpens the distribution.

To prioritize reshaping logits of minority samples and ensure they can exceed the threshold more easily, we introduce an adaptive temperature that adjusts the smoothness based on the individual score of each sample. This adaptivity is achieved by modifying the temperature factor τ in Eq. (12) to a function $\tau_i(\beta)$ on the sample's score, and then the score-based temperature distribution can be expressed as:

$$\begin{aligned} \tau_i(\beta) &= e^{-\beta|\hat{s}_i - \bar{s}|}, \\ \hat{\boldsymbol{y}}_i &= \frac{e^{\boldsymbol{z}_i/\tau_i(\beta)}}{\sum_{j=1}^{B} e^{\boldsymbol{z}_j/\tau_j(\beta)}}, \end{aligned} \tag{13}$$

Here, $|\hat{s}_i - \bar{s}|$ measures the difference between a sample's predicted score $\hat{\boldsymbol{s}}_i$ and the average score. The hyper-parameter $\beta > 0$ controls the steepness of temperature decline as the score difference increases. This exponentially decreasing temperature function prioritizes sharpening for minority samples. As the score difference grows, the temperature approaches 0, resulting in a sharper distribution of the minority. Conversely, majority samples with smaller differences maintain a smoother distribution.

Determining Hyper-parameter by Grid Search. Unlike previous works where the threshold is typically an independent hyper-parameter to derive pseudo-labels, we integrate the threshold and sharpening magnitude as interconnected hyper-parameters. To achieve this, we introduce an additional stage to explore diverse combinations of these two hyper-parameters via grid search. This grid search is guided by MAE between generated and ground-truth labels. The search goal is to pinpoint the specific hyper-parameter combination that yields the most reliable and accurate pseudo-labels.

Initially, we construct a small labeled dataset, which can be sampled randomly from the validation set. We then employ the trained model to assess the samples from this dataset and select pseudo-labels based on the APDS strategy. Following this, a set of pseudo-labeled samples is generated, represented as $D' = \{(\boldsymbol{x}_i, \boldsymbol{y}_i) | \max(\hat{\boldsymbol{y}}_i(\beta)) \geq t\}_{i=1}^{M}$, where $\max(\hat{\boldsymbol{y}}_i(\beta)) \geq t$ denotes the condition for pseudo-label screening. Next, we calculate the labeled score s_i and predicted score $\hat{s}_i$ for each sample in $D^{'}$ (noting that this is based on the formula $s_i = \sum_{j=1}^{C} j \times y_i^j$ and $\hat{s}_i = \sum_{j=1}^{C} j \times \hat{y}_i^j$). By adjusting the magnitude β and the selection threshold t, different pseudo-labels can be generated. Finally, we aim to find the parameters β^* and t^* that minimize the MAE through a grid search method, which can be expressed as:

$$\beta^*, t^* = \arg\min_{\beta, t} \sum_{i=1}^{M} |\hat{s}_i(\beta, t) - s_i| \tag{14}$$

Once this optimal configuration is identified, we leverage it to rigorously filter pseudo-labels with corresponding generated images for subsequent use in the self-training process.

6 Experiments

6.1 Experimental Settings

Datasets. We evaluate the performance of our approach on four representative datasets: AVA [28], AADB [58], TAD66K [30], PARA [29], which are the general, multi-attribute, theme-oriented and personalized aesthetic datasets, respectively, for IAA tasks. The AVA dataset is the most extensive available aesthetic dataset, containing over 250,000 images, and each image is associated with a distribution of scores in a range of 1-10. Similar to [28, 30, 37, 38], we use 235,528 images for model training and 20,000 images for testing. AADB is an aesthetic attribute dataset containing about 10,000 images, each scored from 1 to 5. Following the previous work [58, 61, 78], we use the standard split with 8,500 for training, 1,000 for testing and 500 for validation. The TAD66K dataset contains about 66,000 images covering 47 popular themes, and each image has been annotated with score from 1 to 10 based on dedicated theme evaluation criteria. We use the official train-test split [30, 42], 52,248 for training and 14,079 for testing. PARA is a personalized image aesthetics dataset with rich attributes, which consists of 31,220 images with annotated scores from 1 to 5. The train-test split is the same with the work [29]. During self-training, we used a total of about 50,000 samples (minority and majority), of which AVA, AADB, TAD66K and PARA used 35,000, 5,000, 5,000 and 5,000 samples, respectively. These samples will be open sourced to the community as supplementary datasets.

Evaluation Metrics. We adopt two well-known evaluation metrics, the Pearson Linear Correlation Coefficient (PLCC, $\mathcal{P}$) and the Spearman Rank Correlation Coefficient (SRCC, $\mathcal{S}$) to evaluate performance. While models trained on long-tail datasets often struggle with minority samples, these samples typically make up a small portion of the datasets and thus have a limited impact on both holistic SRCC and PLCC metrics. To further analyze the long-tail issue for labels and scenes, we define two new metrics based on the Mean Absolute Error (MAE) and aforementioned boundary definitions (Sec. 3.2). For the label-level analysis, we sort each dataset by its ground-truth scores and split it into three categories based on percentiles: top 20% as "high", bottom 20% as "low", and the remaining 60% as "medium". Then the MAE for the low, medium, high categories is calculated, denoted as $\mathcal{LL}$, $\mathcal{LM}$, $\mathcal{LH}$, respectively. Similarly, for the scene-level analysis, we calculate the MAE on minority and majority scenes. These metrics are denoted as $\mathcal{SN}$ for minority scenes and $\mathcal{SJ}$ for majority scenes.

Benchmark Models. As shown in Table 2, we conduct a comparative analysis involving 16 models, including 9 SOTA IAA models: NIMA [37], HGCN [39], BIAA [61], TANet [30], MaxViT [41], MUSIQ [40], EAT [42], AesExpert [55], and Q-Align [57], along with 2 Deep Imbalanced Regression (DIR) models [63, 64], which excel at long-tail regression tasks. Additionally, we retrained 3 models—Swin Transformer V2 [62], LLaVA [46], and Q-Instruct [53]—that demonstrated outstanding performance on IAA tasks from other vision domains. To ensure fairness, we used the same Swin Transformer V2 backbone for all DIR methods and for both ELTA 1.0 and ELTA 2.0. Since most of the models have not been tested on all four datasets, we ***retrained them*** using their publicly available code and recommended parameter settings.

Training Details. The loss is categorized into two types: labeled loss L_{s}, derived from the original training set, and unlabeled loss L_{u}, sourced from the pseudo-labeled samples. For labeled loss, since the ground truth in the AVA dataset consists of the score distribution, we use the Earth Mover's Distance (EMD) loss to measure the distance between the ground-truth and the predicted distribution. For the AADB, TAD66K, and PARA datasets, we use mean squared error (MSE) as the labeled loss, as these datasets only provide holistic scores. The training process is optimized using the Adam optimizer, with a batch size of 48. Once the first round of training is completed, model evaluation is performed to generate images and select pseudo-labels. We use these pseudo-labels for only one additional round of self-training. During the self-training process, the loss is calculated as follows:

$$L_{\mathrm{s}} = \frac{1}{B^l}\sum_{i=1}^{B^l} H\left(\boldsymbol{y}_i^l, f\left(\boldsymbol{x}_i^l\right)\right),$$
$$L_{\mathrm{u}} = \frac{1}{B^u}\sum_{i=1}^{B^u} \mathbb{1}\left[\max\left(\hat{\boldsymbol{y}}_i^u(\beta^*)\right) \geq t^*\right] \cdot H\left(\hat{\boldsymbol{y}}_i^u, f\left(\boldsymbol{x}_i^u\right)\right), \tag{15}$$

where superscripts l and u denote labeled and unlabeled data, respectively. H denotes the loss function and $\hat{\boldsymbol{y}}_i^u$ represents the generated pseudo-labels. The indicator function, denoted by $\mathbb{1}[\cdot]$, assigns a value of 1 only when the maximum value of $\hat{\boldsymbol{y}}_i^u$ exceeds the threshold t^*; otherwise, its value is 0.

6.2 Performance Evaluations

In this section, we evaluate the impact of the long-tail issue caused by the label and scene distributions on the model's performance. We compare our method with both the IAA and DIR methods across four datasets to assess the effectiveness of each approach.

Long-tail Issue at Label Level. Table 3 presents performance comparisons with IAA methods. Three key observations emerge from these results: 1) ELTA 2.0 achieves SOTA performance in both $\mathcal{LL}$ and $\mathcal{LH}$ across all datasets, showing significant improvement over ELTA 1.0. This indicates that self-training effectively reduces evaluation errors for the minority. 2)

Table 3: A comparison of 14 models on four representative datasets, evaluating overall performance as well as performance on the majority ($\mathcal{LM}$) and minority ($\mathcal{LL}, \mathcal{LH}$) of the ***label distribution***. All models were implemented using their officially recommended configurations from the open-source code to ensure their best performance. The best result for each metric is bolded in **red**, and the second best result is marked in **blue**.

Dataset	Metric	CNN-based models				Transformer-based models				MLLMs-based models				ELTA	
		NIMA	HGCN	BIAA	TANet	Swin	MaxViT	MUSIQ	EAT	LLaVA	Q-Instruct	AesExpert	Q-Align	1.0	2.0
AVA	$\mathcal{P}\uparrow$	0.636	0.687	0.668	0.765	0.743	0.745	0.738	0.770	0.777	0.778	0.779	**0.817**	0.777	**0.793**
	$\mathcal{S}\uparrow$	0.612	0.665	0.651	0.758	0.735	0.708	0.726	0.759	0.781	0.782	0.784	**0.822**	0.764	**0.810**
	$\mathcal{LL}\downarrow$	0.655	0.675	0.653	0.630	0.616	0.600	0.647	0.490	0.483	0.481	0.480	0.470	**0.438**	**0.412**
	$\mathcal{LM}\downarrow$	0.322	0.321	0.382	**0.237**	0.295	0.317	0.305	0.313	0.312	0.315	0.311	0.309	0.302	**0.286**
	$\mathcal{LH}\downarrow$	0.648	0.660	0.568	0.729	0.513	0.531	0.628	0.433	0.449	0.446	0.437	0.468	**0.426**	**0.410**
AADB	$\mathcal{P}\uparrow$	0.711	0.734	0.733	0.742	0.740	0.748	0.761	0.767	0.769	0.767	0.771	0.770	**0.772**	**0.791**
	$\mathcal{S}\uparrow$	0.700	0.716	0.710	0.749	0.732	0.742	0.751	0.759	0.761	0.762	**0.767**	0.762	0.760	**0.782**
	$\mathcal{LL}\downarrow$	1.450	1.453	1.508	1.394	1.526	1.592	1.447	1.375	1.369	1.370	1.363	1.388	**1.289**	**1.256**
	$\mathcal{LM}\downarrow$	0.874	0.989	0.897	0.846	0.896	**0.782**	0.880	0.828	0.823	0.830	0.821	**0.804**	0.905	0.874
	$\mathcal{LH}\downarrow$	1.527	1.299	1.423	1.355	1.402	1.461	1.159	1.260	1.256	1.255	1.251	1.239	**1.141**	**1.129**
TAD66K	$\mathcal{P}\uparrow$	0.405	0.493	0.431	0.531	0.507	0.513	0.517	**0.546**	0.525	0.524	0.519	0.536	0.539	**0.540**
	$\mathcal{S}\uparrow$	0.390	0.486	0.417	**0.513**	0.478	0.484	0.489	**0.517**	0.508	0.501	0.493	0.506	0.496	0.508
	$\mathcal{LL}\downarrow$	1.851	1.808	1.734	1.598	1.621	1.570	1.627	1.591	1.660	1.622	1.620	1.609	**1.457**	**1.441**
	$\mathcal{LM}\downarrow$	0.812	0.780	0.876	**0.682**	0.793	0.746	0.728	0.782	0.742	0.739	0.727	**0.722**	0.812	0.801
	$\mathcal{LH}\downarrow$	1.690	1.370	1.669	1.651	1.354	1.402	1.460	1.175	1.535	1.531	1.448	1.650	**1.162**	**1.153**
PARA	$\mathcal{P}\uparrow$	0.862	0.881	0.886	0.899	0.925	0.936	0.918	0.940	0.941	0.941	0.937	0.940	**0.943**	**0.946**
	$\mathcal{S}\uparrow$	0.877	0.865	0.858	0.887	0.897	0.902	0.899	0.909	0.907	0.906	0.901	**0.923**	0.912	**0.916**
	$\mathcal{LL}\downarrow$	0.616	0.573	0.469	0.551	0.402	0.383	0.572	0.336	0.330	0.333	0.345	0.340	**0.327**	**0.315**
	$\mathcal{LM}\downarrow$	0.344	0.290	0.328	0.299	0.314	0.282	0.315	0.276	0.272	0.270	0.289	0.279	**0.251**	**0.247**
	$\mathcal{LH}\downarrow$	0.486	0.502	0.503	0.429	0.379	0.276	0.424	**0.256**	0.258	0.260	0.274	0.271	0.290	**0.250**

Table 4: A comparison of 14 models on four representative datasets, evaluating overall performance as well as performance on the majority ($\mathcal{SJ}$) and minority ($\mathcal{SN}$) of the ***scene distribution***. All models were implemented using their officially recommended configurations from the open-source code to ensure their best performance. The best result for each metric is bolded in **red**, and the second best result is marked in **blue**. The holistic performance $\mathcal{S}$ and $\mathcal{P}$ are the same as those in Table 3.

Dataset	Metric	CNN-based models				Transformer-based models				MLLMs-based models				ELTA	
		NIMA	HGCN	BIAA	TANet	Swin	MaxViT	MUSIQ	EAT	LLaVA	Q-Instruct	AesExpert	Q-Align	1.0	2.0
AVA	$\mathcal{SN}\downarrow$	0.528	0.520	0.511	0.518	0.435	0.457	0.493	0.395	0.412	0.407	0.403	0.419	**0.390**	**0.339**
	$\mathcal{SJ}\downarrow$	0.456	0.443	0.421	0.430	0.359	0.380	0.428	0.339	0.339	0.347	0.341	0.351	**0.331**	**0.316**
AADB	$\mathcal{SN}\downarrow$	1.285	1.259	1.246	1.189	1.243	1.205	**1.096**	1.127	1.115	1.120	1.111	1.109	1.108	**0.926**
	$\mathcal{SJ}\downarrow$	1.094	1.011	1.055	1.002	1.033	1.002	0.932	0.941	0.957	0.950	0.953	0.916	**0.909**	**0.877**
TAD66K	$\mathcal{SN}\downarrow$	1.348	1.156	1.359	1.265	1.154	1.154	1.185	1.066	1.229	1.221	1.171	1.272	**1.059**	**0.902**
	$\mathcal{SJ}\downarrow$	1.137	0.985	1.166	1.051	0.976	0.963	0.982	**0.875**	1.018	1.031	0.999	1.077	0.898	**0.877**
PARA	$\mathcal{SN}\downarrow$	0.449	0.424	0.445	0.388	0.375	0.297	0.395	0.285	0.287	**0.283**	0.301	0.301	0.290	**0.279**
	$\mathcal{SJ}\downarrow$	0.378	0.359	0.379	0.334	0.311	0.256	0.336	**0.240**	**0.237**	0.243	0.258	0.244	0.245	0.246

Although some methods outperform ours in holistic metrics (PLCC and SRCC) on certain datasets, they lag in minority performance. This highlights their limitations in handling long-tail data, even for MLLMs-based methods pretrained on large-scale data. This discrepancy also exposes a critical limitation of the popular PLCC and SRCC metrics: they fail to reflect the long-tail issue in IAA models. 3) Our method slightly sacrifices majority class performance, a common trade-off in long-tail solutions [79].

Long-tail Issue at Scene Level. Table 4 presents performance comparisons with IAA methods at the scene level. Our proposed model, ELTA 2.0, consistently achieves SOTA performance on minority scenes across

Table 5: Comparing our ELTA with 2 DIR methods, L&FDS [63] and Ranksim [64].

Dataset	Metric	L&FDS	RankSim	Base.	ELTA
AVA	$\mathcal{SN}\downarrow$	0.375	**0.372**	0.435	**0.339**
	$\mathcal{SJ}\downarrow$	0.428	0.431	**0.359**	**0.316**
	$\mathcal{LC}\downarrow$	0.588	**0.542**	0.616	**0.412**
	$\mathcal{LM}\downarrow$	0.303	0.297	**0.295**	**0.286**
	$\mathcal{LH}\downarrow$	0.509	**0.485**	0.513	**0.410**
AADB	$\mathcal{SN}\downarrow$	1.066	**1.008**	1.243	**0.926**
	$\mathcal{SJ}\downarrow$	1.218	1.157	**1.033**	**0.877**
	$\mathcal{LC}\downarrow$	**1.473**	1.537	1.526	**1.256**
	$\mathcal{LM}\downarrow$	0.888	**0.881**	0.896	**0.874**
	$\mathcal{LH}\downarrow$	1.407	**1.295**	1.402	**1.129**
TAD66K	$\mathcal{SN}\downarrow$	1.001	**0.988**	1.154	**0.902**
	$\mathcal{SJ}\downarrow$	1.144	1.139	**0.976**	**0.877**
	$\mathcal{LC}\downarrow$	1.611	**1.560**	1.621	**1.441**
	$\mathcal{LM}\downarrow$	**0.787**	0.796	**0.793**	0.801
	$\mathcal{LH}\downarrow$	1.369	**1.343**	1.354	**1.153**
PARA	$\mathcal{SN}\downarrow$	**0.312**	0.323	0.375	**0.279**
	$\mathcal{SJ}\downarrow$	0.344	0.366	**0.311**	**0.246**
	$\mathcal{LC}\downarrow$	0.371	**0.363**	0.402	**0.315**
	$\mathcal{LM}\downarrow$	**0.301**	0.322	0.314	**0.247**
	$\mathcal{LH}\downarrow$	0.360	**0.345**	0.379	**0.250**

Table 6: Cross-dataset evaluations (SRCC↑) across training and testing datasets.

Cross dataset	Training dataset	Testing dataset AVA	TAD66K	PARA
RankSim	AVA	0.753	0.415	0.694
	TAD66K	0.613	0.487	0.728
	PARA	0.660	0.381	0.906
ELTA 1.0	AVA	0.764	0.443	0.706
	TAD66K	0.701	0.496	0.795
	PARA	0.667	0.406	0.912
ELTA 2.0	AVA	0.810	0.488	0.797
	TAD66K	0.724	0.508	0.831
	PARA	0.715	0.434	0.916

Table 7: Cross-architecture evaluations are conducted to enhance other IAA methods, resulting in improved results on the **AVA** dataset.

Model	+ Module TFA	FLSA	APDS	Metric $\mathcal{P}\uparrow$	$\mathcal{S}\uparrow$	$\mathcal{LC}\downarrow$	$\mathcal{LM}\downarrow$	$\mathcal{LH}\downarrow$
NIMA				0.636	0.612	0.655	**0.322**	0.648
	✓	✓		0.649	0.631	0.614	0.338	0.628
	✓	✓	✓	**0.658**	**0.640**	**0.593**	0.340	**0.601**
HGCN				0.687	0.665	0.675	**0.321**	0.660
	✓	✓		0.700	0.675	0.582	0.343	0.593
	✓	✓	✓	**0.714**	**0.693**	**0.564**	0.329	**0.575**
BIAA				0.668	0.651	0.653	0.382	0.568
	✓	✓		0.682	0.675	0.596	0.390	0.514
	✓	✓	✓	**0.699**	**0.687**	**0.544**	**0.381**	**0.497**
MUSIQ				0.738	0.726	0.647	**0.305**	0.628
	✓	✓		0.748	0.734	0.603	0.311	0.562
	✓	✓	✓	**0.761**	**0.745**	**0.588**	0.327	**0.482**
MaxViT				0.745	0.708	0.600	**0.317**	0.531
	✓	✓		0.750	0.728	0.557	0.340	**0.426**
	✓	✓	✓	**0.759**	**0.742**	**0.532**	0.333	0.428
TANet				0.765	0.758	0.630	**0.237**	0.729
	✓	✓		0.772	0.767	0.591	0.244	0.633
	✓	✓	✓	**0.779**	**0.771**	**0.562**	0.251	**0.585**
EAT				0.770	0.759	0.490	0.313	0.433
	✓	✓		0.777	0.765	**0.450**	0.310	0.426
	✓	✓	✓	**0.780**	**0.768**	**0.450**	**0.307**	**0.413**

all four datasets, demonstrating its superior capability in mitigating data imbalance. Furthermore, the long-tail effect at the scene level causes a smaller performance gap between $\mathcal{SJ}$ and $\mathcal{SN}$ compared to imbalances at the label level. Despite this, the clear struggle of existing methods on minority scenes underscores their fundamental inadequacy in handling the long-tail issue.

Comparison with DIR methods. The results of the comparison with DIR methods are shown in Table 5. Our approach outperforms the methods presented in [63] and [64]. Specially, it excels in prediction for minority samples, indicating a significant improvement in the IAA area. This demonstrates that our ELTA is more suitable for IAA tasks compared to some general methods used to solve long-tail problems.

Cross-dataset Evaluations. Long-tail adaptation methods can easily overfit to dataset-specific distributions. To assess generalization, we conduct cross-dataset evaluation among AVA, TAD66K, and PARA. Because these datasets use different label ranges, rank-based correlations are preferable to absolute-error metrics (MAE), we therefore report SRCC.

As shown in Table 6, ELTA-2.0 achieves the highest SRCC for each pair. The cross-dataset average reaches 0.665, exceeding ELTA-1.0 (0.620) and RankSim (0.582). The largest margin occurs when the target dataset is TAD66K: ELTA-2.0 attains an average SRCC of 0.477, a +11.5% gain over RankSim (0.428) and +6.3% over ELTA-1.0. We attribute these gains to ELTA-2.0's scene-level long-tail handling, which produces more accurate predictions on minority scenes and, in turn, stronger transfer across datasets.

Table 8: Ablation study of the self-training strategy on the **AVA** dataset.

Model	+ Self-training	Metric $\mathcal{SN}\downarrow$	$\mathcal{SJ}\downarrow$
NIMA		0.528	0.456
	✓	**0.503**	**0.445**
MUSIQ		0.493	0.428
	✓	**0.471**	**0.420**
TANet		0.518	0.430
	✓	**0.496**	**0.425**
EAT		0.395	0.339
	✓	**0.366**	**0.337**
LLaVA		0.412	0.339
	✓	**0.389**	**0.328**

Table 9: Ablation of different modules on the **AVA** dataset.

Module TFA	FLSA	APDS	Metric $\mathcal{P}\uparrow$	$\mathcal{S}\uparrow$	$\mathcal{LL}\downarrow$	$\mathcal{LM}\downarrow$	$\mathcal{LH}\downarrow$
			0.743	0.735	0.616	0.295	0.513
✓			0.749	0.740	0.518	0.323	0.474
	✓		0.758	0.752	0.592	0.287	0.520
		✓	0.760	0.749	0.575	0.286	0.480
✓	✓		0.764	0.748	0.484	0.324	0.431
✓		✓	0.762	0.756	0.461	0.314	0.445
	✓	✓	0.771	0.757	0.547	**0.286**	0.463
✓	✓	✓	**0.777**	**0.764**	**0.438**	0.302	**0.426**

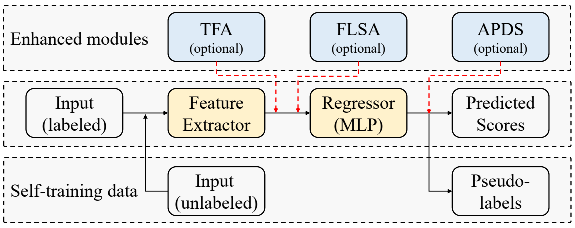


Fig. 8: Example of plugging our modules into other methods.

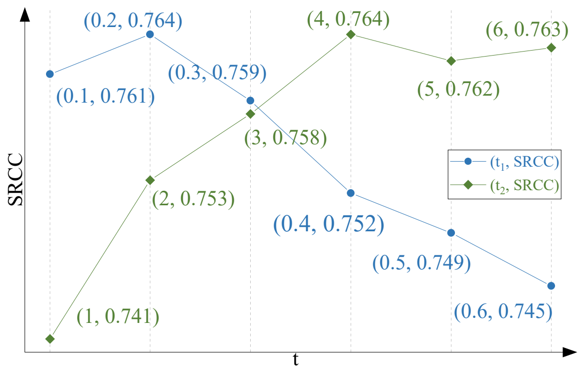


Fig. 9: Effect of temperature hyper-parameters (τ_1 and τ_2) on SRCC.

6.3 Enhancing other Methods

By Plugging Three Modules. To demonstrate the effectiveness of our proposed ELTA and its seamless integration into existing methods, ELTA is plugged into the other seven IAA methods. As illustrated in Fig. 8, we integrate FLSA and TFA as an optimization module for representation learning; following this, we incorporate self-training with the APDS module to enhance performance. Results are detailed in Table 7. The data indicates improvements in the PLCC, SRCC, and MAE in both low and high segments.

By Self-training on our Supplementary Dataset. As shown in Table 8, we used data from our AG-T2I model to create a supplementary dataset (Sec. 6.1). When this dataset was used to train other models, their performance at the scene level markedly improved, highlighting the effectiveness of our self-training strategy.

6.4 Ablations and Analysis

Effectiveness of the Modules. To assess the effectiveness of the modules, we create eight combinations by selecting whether to use each of the three modules (based on Swin Transformer V2). The results on the AVA dataset are shown in Table 9. We observe that the TFA module has a relatively significant effect on reducing minority errors, while FLSA and APDS demonstrate balanced improvements across all metrics. For example, removing the TFA module (row 7) leads to an increase in MAE by 24.9% for the low segment and 8.7% for the high segment, in contrast to the configuration with all three modules active (row 8).

Ablations on Sampling Strategy in TFA. Two ablation tests are carried out to validate the necessity of the proposed sampling strategy in the TFA module. The first employs a random sampling strategy. The results indicated a 4.6% decrease in both PLCC and SRCC. By examining the labels associated with the mixed features, we find that the augmentation occurs mainly in the majority rather than in the minority. In the second test, we set thresholds to distinguish between majority and minority, allowing only minority features defined in this way to be mixed. Despite experimenting with various thresholds, the optimal results achieved were 0.755 and 0.747, still below the 0.777 and 0.764 achieved with our strategy.

Ablations on Hyper-parameters. In Eq. (2) and (3), two temperature hyper-parameters are used to adjust the sampling probability distribution. They determine how much we favor the minority when selecting

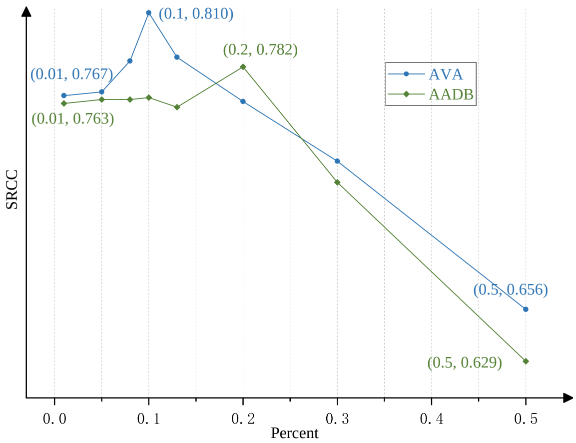


Fig. 10: Effect of sample numbers in self-training, the horizontal axis shows the percentage of minority sample numbers from the supplementary dataset relative to the total number of samples in the AVA or AADB dataset.

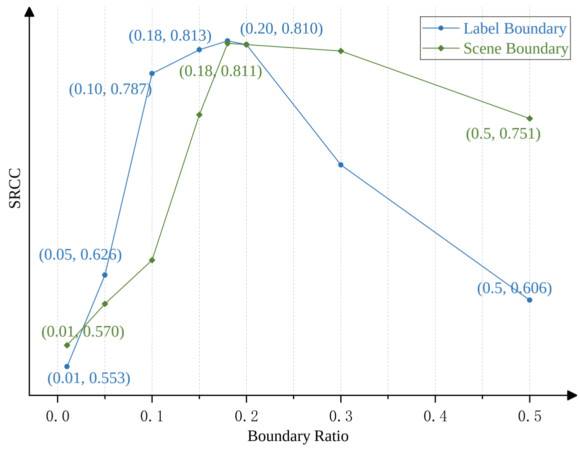


Fig. 11: Effect of boundary settings for majority and minority labels and scenes. Performance degrades when the boundary is either too strict or too loose on the AVA dataset.

Table 10: Comparisons of feature-level augmentation and pixel-level augmentation on **AVA**.

Model	$\mathcal{P}\uparrow$	$\mathcal{S}\uparrow$	$\mathcal{LL}\downarrow$	$\mathcal{LM}\downarrow$	$\mathcal{LH}\downarrow$
TFA(pixel)	0.731	0.719	0.597	0.368	0.541
TFA(feature)	0.749	0.740	0.518	0.323	0.474
TFA(pixel)+FLSA+APDS	0.752	0.741	0.487	0.340	0.485
TFA(feature)+FLSA+APDS	**0.777**	**0.764**	**0.438**	**0.302**	**0.426**

samples. To quantitatively assess how these hyper-parameters affect results, we conduct tests with various hyper-parameter values. The outcomes on AVA are shown in Fig. 9. Notably, the SRCC metric reaches its peak with τ_1 set at 0.2 and τ_2 at 4.0.

Ablations on the Sample Numbers of Self-Training. To investigate the impact of adding minority sample numbers from the supplementary dataset for self-training, we analyzed performance changes by gradually increasing the number of minority samples, as shown in Fig. 10. The results indicate that when the sample size is too small, the model's performance is similar to the ELTA 1.0 model, which relies solely on feature dimension augmentation. Performance peaks when the minority sample proportion reaches approximately 10% (AVA) and 20% (AADB). However, when the number of minority samples is excessively high, the model's performance declines sharply.

Ablations on Augmentation Dimension. In Section 4.1, we pointed out that, ***without changing the label***, feature-dimension enhancement is a more suitable augmentation approach for IAA. Here, we present some quantitative experimental results. We maintained the same sample selection strategy for mixup but shifted the mixup process from the feature dimension to the pixel dimension of the original images. The results in Table 10 indicate a noticeable decline in performance, regardless of whether we applied the TFA module individually or all modules together.

Ablations on Boundary Definitions. In Section 3.2, we define the boundaries for the majority and minority labels and scenes, with the default boundary settings empirically set to 20%. Since this choice is based on prior experience, we further investigate how varying the boundary settings affects model performance. As shown in Fig. 11, although certain boundary settings yield slightly better results than the 20% setting, the overall trend indicates a performance drop when the boundary becomes either too strict or too loose. A lower ratio results in overly strict boundaries, causing very few minority labels or scenes to be considered and thus limiting the granularity of the enhancement. On the other hand, a higher ratio leads to overly loose boundaries, which further exacerbates the long-tailed problem.

6.5 Visualization.

Predictions for Images. To better understand the impact of the long-tail issue on IAA models, we present a qualitative analysis. Fig. 12 displays examples from the AVA and TAD66K datasets, comparing a baseline model against our proposed ELTA model. For a fair comparison, the baseline uses the same Swin Transformer V2 backbone as ELTA.

At the label level, the long-tail distribution increases prediction errors for minority samples. This issue also forces the model's scores into a ***conservative and narrow range***, typically between 4.0 and 6.5. Consequently, in practical applications, the model struggles to distinguish between images of very high or very low aesthetic quality. This issue extends to the scene level, where the long-tail distribution degrades performance on minority scenes and amplifies prediction

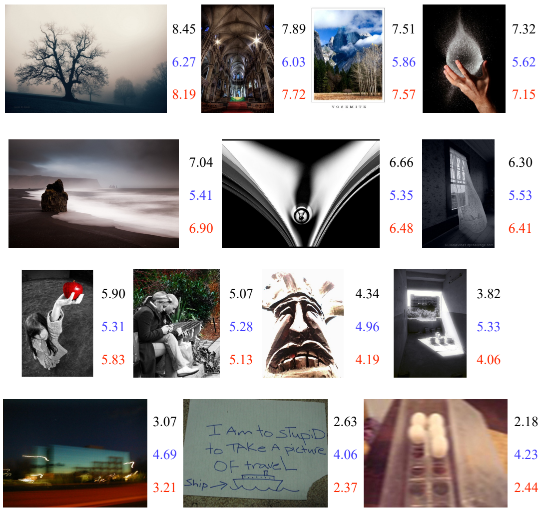


Fig. 12: The images from the AVA and TAD66K datasets are visualized with their corresponding ground-truth scores, alongside the predicted scores from both the baseline model (marked in blue) and ELTA (marked in red).

errors. For instance, the sample in the first row belongs to a minority group in terms of both its label and scene. As a result, the baseline model exhibits a much larger prediction error for this image compared to others. In contrast, our proposed ELTA model effectively mitigates this long-tail issue, demonstrating more accurate and discerning predictions.

Style and Aesthetic Control Capability of AG-T2I. To qualitatively illustrate the AG-T2I model's control ability, we provide examples in Fig. 13. Guided by instructions, the model mimics the style of the source image, generating images that match the specified scene and anchor aesthetic score. The actual aesthetic quality of the images, as predicted by ELTA, varies slightly around the anchor aesthetic score. To further validate this capability, for each anchor score of 3, 5, and 7, we generated 100 images and analyzed the distribution of their predicted scores. The results, shown in Fig. 14, demonstrate a close alignment between the predicted and anchor scores.

To quantitatively evaluate this capability, the predicted scores across these image sets yield a low MAE of 0.43 relative to the anchor scores. Additionally, the generated images' style features achieve a high cosine similarity of 0.88 with the average style embeddings of stylistically matched the source image. These results collectively confirm that AG-T2I of ELTA 2.0 enables effective and reliable joint control of both style and aesthetic quality.

6.6 Complexity Analysis

We present the complexity analysis, including the number of training parameters and inference time measured on 4 × 48GB NVIDIA A6000 GPUs. As shown in Fig.

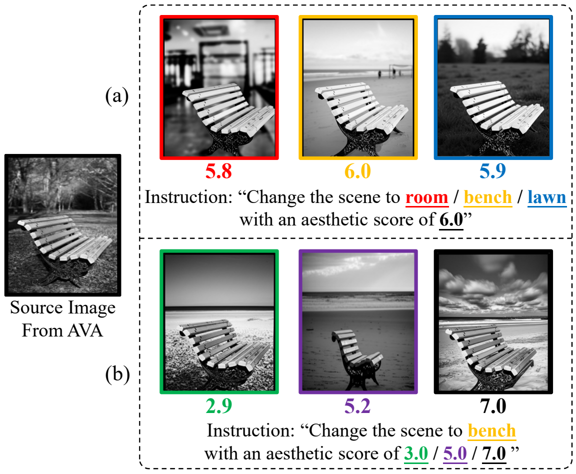


Fig. 13: The AG-T2I model generates images based on instructions to control style and aesthetic quality, with corresponding aesthetic scores predicted by ELTA displayed below each image.

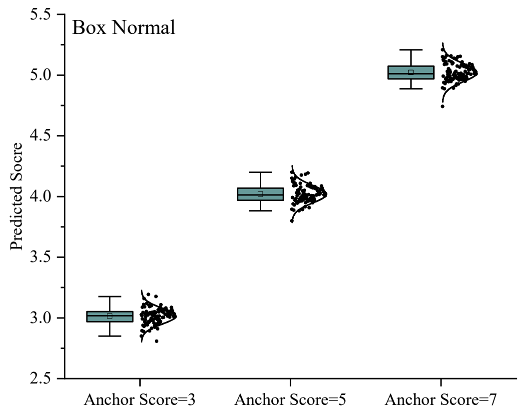


Fig. 14: Distribution of predicted aesthetic scores versus anchor scores in instructions.

4, the overall model complexity is primarily influenced by two components: the Feature Extractor, which uses a SwinV2-Base backbone with 87.0M parameters, and the AG-T2I module, where we fine-tune the Stable Diffusion 1.5 model, adding an additional 0.98B parameters. The inference time for predicting the aesthetic score of a single image is approximately 325 ms.

6.7 Failure Cases

We identify two failure cases where the model yields limited performance improvement. The first occurs when ELTA augments existing data at the feature level. Specifically, we observe that if the minority subset contains visually similar images with significantly different aesthetic scores, this may introduce label noise and disrupt the alignment between features and labels. As a result, the effectiveness of ELTA in enhancing tail performance becomes less pronounced.

The second case arises when ELTA generates new data at the scene level. In this setting, the style control capability partially depends on the accuracy of a pre-trained scene classifier. For minority samples with ambiguous or poorly defined scene categories, the classifier may fail to provide reliable editing instructions, thereby reducing the quality and relevance of the generated data.

7 Discussion AND FUTURE DIRECTIONS

From the evaluation, we observe that while many SOTA IAA models, including MLLMs, achieve high overall performance on holistic metrics like PLCC and SRCC, their performance degrades significantly on minority data. This indicates that high holistic metrics can mask critical weaknesses in handling real-world imbalanced data. The success of our ELTA framework in improving minority performance, as measured by metrics like $\mathcal{SN}$, $\mathcal{LL}$, $\mathcal{LH}$, underscores that explicitly addressing the long-tail issue is crucial for developing robust and fair IAA applications. This finding prompts a deeper discussion on several key issue and future research directions in the field.

Generalizability *vs.* Specialization. A key debate in AI is whether massive, general-purpose models can outperform smaller, specialized ones. Our findings show that even large MLLMs, pre-trained on vast datasets, struggle with the specific long-tail issue of IAA. In contrast, ELTA demonstrates superior performance on minority data. This suggests that the nuanced and subjective nature of aesthetics may benefit more from specialized approaches rather than relying solely on the broad knowledge of general models. A future direction is to explore a hybrid approach: how can we efficiently inject specialized aesthetic knowledge and long-tail awareness into MLLMs, combining the breadth of general pre-training with the depth of specialized fine-tuning?

Subjectivity and Controlled Data Generation. A core challenge in IAA is the inherent subjectivity and ambiguity in human-provided labels. Our proposed generative model, AG-T2I, addresses the long-tail issue by creating new images with specific, controlled aesthetic scores. While this balances the data distribution, it raises an important question: does generating data for discrete "anchor" scores fully capture the continuous and often ambiguous spectrum of human aesthetic perception? The generated data is "cleaner" than the original noisy dataset, which is beneficial for training, but may not reflect the full complexity of aesthetic judgment. Future research could focus on generative models that learn to replicate the distribution of human opinions for an image, rather than a single mean score, thereby generating data that better reflects real-world label ambiguity.

Fairness in Aesthetic Evaluation. Our work highlights that traditional metrics like PLCC and SRCC do not adequately capture a model's performance across the entire data distribution, often hiding biases against minority groups. We introduced more granular metrics to enable a fairer assessment. This is a critical step, but more work is needed. The goal should be to develop unified metrics that intrinsically balance overall accuracy with fairness, penalizing models that perform poorly on underrepresented data. How to mathematically formulate such a metric remains an important open question for the community.

Weaknesses. Despite the strong performance of the ELTA framework, it has several limitations. First, in line with other long-tail methods, ELTA's significant gains on minority data sometimes come at the cost of a minor performance decrease on majority data. Second, the full ELTA 2.0 pipeline, which involves fine-tuning a T2I model, generating data, and performing self-training with hyper-parameter search, is computationally intensive and more complex than traditional end-to-end approaches. Finally, our methodology's effectiveness on scene balancing relies on an external pre-trained scene classification model, and the generative quality depends on a pre-trained diffusion model backbone, making our framework's performance contingent on these upstream models.

Potential Applications. The ability of ELTA to fairly and robustly assess aesthetics opens up several impactful applications:

1) Fair Automated Image Curation. In applications like online photo contests or automated gallery curation, current AI systems may be biased towards popular subjects and styles. An ELTA-powered system could provide a more equitable platform, recognizing high-quality work in minority categories that might otherwise be overlooked.

2) Enhanced Personalized Recommendation. For personalized IAA tasks (e.g., on the PARA dataset), users often have unique tastes that represent a "minority" preference relative to the general population. By effectively handling the long-tail issue, ELTA can better cater to these niche preferences, providing recommendations that are truly personalized instead of regressing to the mean.

3) Aesthetics-Guided Creative Tools. The AG-T2I model can be repurposed as an intelligent assistant for photographers and digital artists. The model could not only score an image but also provide generative feedback, showing users specific edits or variations that would align the image with a desired aesthetic target (e.g., "make this photo more dramatic" or "give this a vintage feel with an aesthetic score of 7.5").

8 Conclusion

This paper reveals the long-tail distribution in the IAA datasets, its specificity, and the consequent model bias. To address this issue, we propose ELTA, a comprehensive framework that operates on two levels to alleviate the long-tail issue. To foster further research, we developed a benchmark evaluating 16 methods, where our ELTA model achieves SOTA performance across all tests. Additionally, we released a supplementary dataset containing over 50,000 images to address imbalances in existing datasets, along with a toolkit for long-tail analysis, automated data generation, and annotation.

We aspire for our work to make a valuable contribution to the ongoing IAA research on the long-tail issue within the community. However, despite notable progress, some challenges remain unresolved. Future work will focus on addressing these limitations to further enhance performance.

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